AGENTIC AI ZERO TO HERO · #1 · CHEAT SHEET

What Is Agentic AI? 70 Years of AI in 7 Eras

youtube.com/@PavanElthepu
github.com/pelthepu/agentic-ai-zero-to-hero

#	ERA	WHEN	BIG IDEA	WHAT BROKE	"PLAN A 5-DAY JAPAN TRIP"
1	**The Question**	1950	Alan Turing asks "Can machines think?" and proposes the Turing Test	Only an idea, no working system	—
2	**Rule-based AI**	1960s	People write IF-THEN rules by hand	Needs too many rules. No real understanding	Always Tokyo, Kyoto, Mount Fuji
3	**AI Winter**	1970s	Big promises, slow progress, funding cut	Not enough data or computing power	—
4	**Machine Learning**	1980s–2010s	Machines learn patterns from data. Deep learning (neural networks) takes off around 2012	One narrow task at a time	The most popular plan from past trips
5	**LLMs & GenAI**	2018+	Models trained on huge amounts of text understand and create language	Only answers. Can't take action	A natural plan with the $2,000 budget split
6	**AI Agents**	Recent	An LLM uses tools (APIs) to finish one task	Fixed goal, fixed tools, fixed workflow	Checks live prices: a plan for $1,850
7	**Agentic AI**	Now	Owns the whole goal: plans, uses many tools, remembers, adapts until done	—	Plans, books, and rebooks after a cancelled flight

UNDER THE HOOD (FOR DEVELOPERS)

```
while not done:
    step = llm.decide(goal, history, tools)  # pick the next action
    result = run(step)                       # your code calls the tool
    history.append(result)                   # feed the result back
```

An agent is a loop. We write this loop in Python in lesson 3.

AGENTIC AI IN ONE LINE

An AI system that pursues a goal over many steps. It reasons, plans, uses tools, remembers context and adapts until the goal is done.

Chatbot
Answers when you ask. Stops.

Agentic AI
Notices, acts, and keeps going until the goal is met.

THE AI FAMILY

AI ⊃ Machine Learning ⊃ Deep Learning ⊃ Generative AI

Generative AI creates new content: LLMs make text, and other models make images, music and video.

Next: Generative AI vs AI Agents vs Agentic AI, the exact difference.